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Modular conveyor belt with dedicated access assembly

Abstract

A dedicated access assembly for a conveyor belt forms a dedicated opening and closing point for the conveyor belt. The dedicated access assembly includes a female portion having one or more open hooks and a male portion including a hinge rod extending between linking elements. The open hook disengages from and engages the hinge rod to selectively open and close the conveyor belt. A latching mechanism can selectively lock the dedicated access assembly.

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Background/Summary

RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 17/789,367, filed Jun. 27, 2022, which is a 371 of PCT/US2021/012152, filed on Jan. 5, 2021, which in turn claims priority to U.S. Provisional Patent Application Ser. No. 62/957,441, filed Jan. 6, 2020 and entitled “Modular Conveyor Belt with Master Link Assembly”, the contents of which are herein incorporated by reference.

FIELD OF THE INVENTION

(1) The present invention relates to the field of power-driven conveyors. More particularly, the present invention relates to a modular conveyor belt with a dedicated access assembly for easily separating and reconnecting the modular conveyor belt at a dedicated location.

BACKGROUND OF THE INVENTION

(2) Modular plastic conveyor belts are widely used in various industries to convey products. Modular plastic conveyor belts are constructed of a series of rows of linked belt modules forming an endless loop. Hinge elements along opposite ends of each row interleave with hinge elements of consecutive rows. A hinge rod inserted in the interleaved hinge elements connects the rows together at hinge joints into an endless conveyor belt loop, allowing articulation of the modules relative to each other.

(3) Dedicated access assemblies can be used in certain locations throughout the length of a conveyor belt to allow an operator to easily separate the belt, make repairs or remove sections, and then reconnect the belt without the need for replacement components or special tools.

SUMMARY OF THE INVENTION

(4) A dedicated access assembly for a conveyor belt comprises female and male components capable of unhooking and re-hooking to each other. The female component includes at least one open hook that engages a hinge rod in the male component. The dedicated access assembly facilitates opening and rejoining of the conveyor belt at one or more dedicated locations.

(5) According to one aspect, a module for a conveyor belt comprises a body extending in height from a top conveying surface to an opposite bottom surface, longitudinally from a first end to a second end and laterally from a first side to a second side, a first set of hinge elements extending from the first end of the body, an open hook extending from the second end of the body for locking the module onto a hinge rod and a recess aligned with the open hook and extending from the second end of the body to form a pocket for trapping the hinge rod.

(6) According to another aspect, a module for a conveyor belt comprises a body extending in height from a top conveying surface to an opposite bottom surface, longitudinally from a first end to a second end and laterally from a first side to a second side, a first set of hinge elements extending from the first end of the body, an open hook extending from the second end of the body for locking the module onto a hinge rod and a latching mechanism for locking the open hook onto the hinge rod. The latching mechanism includes a tip for locking the hinge rod within the open hook and a biasing mechanism for biasing the tip into a locked position.

(7) According to another aspect, a module for a conveyor belt comprises a body extending in height

from a top conveying surface to an opposite bottom surface, longitudinally from a first end to a second end and laterally from a first side to a second side, a first set of hinge elements extending from the first end of the body, a pocket extending from the second end of the body for trapping a hinge rod and a latching mechanism for locking the hinge rod in the pocket. The latching mechanism including a tip for locking the hinge rod within the pocket and a biasing mechanism for biasing the tip into a locked position.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) FIG. 1 is an isometric top view of a portion of a modular conveyor belt employing a dedicated access assembly according to one embodiment;
- (2) FIG. 2 is an isometric top view of the portion of FIG. 1 during opening of the conveyor belt using the dedicated access assembly;
- (3) FIG. 3 is an isometric top view of the portion of FIG. 1 after separation of the dedicated access assembly components;
- (4) FIG. 4 is an isometric bottom view of the portion of the modular conveyor belt of FIG. 1;
- (5) FIG. 5 is a bottom view of the portion of FIG. 2;
- (6) FIG. 6 is a bottom view of the portion of FIG. 3;
- (7) FIG. 7 is an isometric view of the dedicated access assembly of FIG. 1 during disassembly;
- (8) FIG. 8 is an isometric bottom view of the dedicated access assembly of FIG. 7 in an assembled position;
- (9) FIG. 9 is a detailed view of a side edge of the assembly of FIG. 8;
- (10) FIG. 10 is an isometric bottom view of an interior male portion of the dedicated access assembly of FIG. 8;
- (11) FIG. 11 is a detailed side view of an open hook of the female portion of the dedicated access assembly of FIG. 8;
- (12) FIG. 12 is an isometric detailed view of a series of open hooks of the female portion of the dedicated access assembly of FIG. 8;
- (13) FIG. 13 is a detailed view of the side edge of FIG. 9 during disassembly of the dedicated access assembly;
- (14) FIG. 14 is a side view of the dedicated access assembly of FIG. 8 during disassembly;
- (15) FIG. 15 is a side view of the dedicated access assembly of FIG. 8 after unlatching the open hooks;
- (16) FIG. 16 is a detailed isometric view of the side edge of FIG. 9 after separation of the male and female portions;
- (17) FIG. 17 is an exploded view of the bottom of the female portion of a dedicated access assembly showing alignment tabs used to align the components of the female portion;
- (18) FIG. 18 is a detailed view of the alignment tabs of FIG. 17;
- (19) FIG. 19 is another exploded view of the bottom of the female portion of FIG. 17 showing the alignment tabs used to align the components;
- (20) FIG. 20 is a detailed view of the alignment tabs of FIG. 19;
- (21) FIG. 21 is a bottom view of a dedicated access assembly including a latching mechanism according to another embodiment;
- (22) FIG. 22 is an exploded view of the dedicated access assembly of FIG. 21;
- (23) FIG. 23 is an isometric view of the latching mechanism of FIG. 21 in a natural, locking position;
- (24) FIG. 24 shows the latching mechanism of FIG. 23 in a depressed, open position;
- (25) FIG. 25 shows a side edge of the dedicated access assembly of FIG. 21 in a locked position;

- (26) FIG. **26** shows another side edge of the dedicated access assembly of FIG. **21** in the locked position;
- (27) FIG. **27** shows the side edge of FIG. **25** in an unlocked position;
- (28) FIG. **28** shows the side edge of FIG. **26** in an unlocked position;
- (29) FIG. **29** is a side view of the dedicated access assembly of FIG. **21** in an openable position;
- (30) FIG. **30** is a cross-sectional view of the assembly of FIG. **29**;
- (31) FIG. **31** is a side view of the dedicated access assembly of FIG. **29** after unlocking the female portion from the male portion;
- (32) FIG. **32** is a cross-sectional view of the assembly of FIG. **31**;
- (33) FIG. **33** is a bottom view of a dedicated access assembly including a latching mechanism and alignment tabs according to another embodiment;
- (34) FIG. **34** is a bottom view of a dedicated access assembly including a latching mechanism according to another embodiment;
- (35) FIG. **35** is an exploded view of the dedicated access assembly of FIG. **34**;
- (36) FIG. **36** is an isometric view of the latching mechanism of FIG. **34**;
- (37) FIG. **37** is an isometric bottom view of a portion of a modular conveyor belt employing a dedicated access assembly including an openable edge portion;
- (38) FIG. **38** is an isometric bottom view of the portion of FIG. **37** during opening of the openable edge portion to allow opening of the conveyor belt;
- (39) FIG. **39** is a detailed bottom view of the dedicated access assembly of FIG. **37**;
- (40) FIG. **40** is an exploded bottom view of the dedicated access assembly of FIG. **39**;
- (41) FIG. **41** is an exploded top view of the dedicated access assembly of FIG. **39**;
- (42) FIG. **42** is an isometric view of the latching mechanism of the dedicated access assembly of FIG. **39**;
- (43) FIG. **43** shows the latching mechanism and openable end portion of FIG. **39** in a locked position relative to a hinge rod;
- (44) FIG. **44** shows the latching mechanism and openable end portion of FIG. **43** in an unlocked position;
- (45) FIG. **45** is an isometric view of another embodiment of a dedicated access assembly;
- (46) FIG. **46** is an isometric view of the openable end portion of the dedicated access assembly of FIG. **45**.

DETAILED DESCRIPTION

- (47) A modular conveyor belt includes a dedicated access assembly for easily separating and joining the belt at a dedicated location for cleaning, repair or removal. The invention will be described below relative to certain illustrative embodiments, though those skilled in the art will recognize that the invention is not limited to the illustrative embodiments.
- (48) Referring to FIGS. **1-6**, an illustrative portion of a modular conveyor belt **100** comprises a series of hingedly connected standard conveyor belt modules **101** forming a top conveying surface **110** and having ends joined together to form a continuous loop that is driven by a drive, such as a sprocket. The modular conveyor belt **100** is made up of a series of rows, with each row consisting of multiple modules arranged side-by-side, typically in a brick-lay pattern from row to row, across the width of the belt. Alternatively, a row can comprise a single conveyor belt module defining the width of the belt. Examples of suitable modular conveyor belts are those manufactured and sold, for example, by Intralox, LLC of Harahan, LA., USA. The belt modules are conventionally made by injection-molding thermoplastic materials such as polypropylene, polyethylene, acetal, nylon, or composite polymers, to name a few, though other materials may be used.
- (49) Each standard conveyor belt module **101** comprises a body extending in thickness from a top conveying surface **110** to an opposite surface, which may include a drive element, such as a drive bar **112**, for engaging a conveyor drive, such as a sprocket. The drive element can alternatively be located on the sides of the modules or in another suitable location, or comprise an opening or

pocket in the module configured to engage a tooth on the drive. The standard module **101** extends in width from a first side to a second side and in length in a direction of belt travel from a first end to a second end. A series of linking elements, shown as spaced-apart hinge elements **120**, extend longitudinally outward from each end of the standard module body. The hinge elements **120** on one end of the module body are laterally offset from the hinge elements **120** at the other end. The gaps between laterally consecutive hinge elements are sized to receive the hinge elements of modules in an adjacent row of modules in the conveyor belt, such that the openings in the hinge elements align to form a hinge passageway extending along a hinge axis. Hinge rods **130** extending laterally through the hinge passageways formed by the aligned openings in the interleaved hinge elements connect the standard modules **101** to each other.

(50) The hinge elements **120** allow articulation of the modules relative to each other about the hinge rods **130**, enabling the conveyor belt to conform to the sprocket or other belt-guiding devices. The distance between consecutive hinge axes is defined as the belt pitch P , which is also the distance between consecutive drive elements **112**. Each illustrative standard conveyor belt module **101** includes articulating hinge elements **120** at each end.

(51) The modular conveyor belt **100** conveys articles atop the conveying surfaces **110** of the modules along an upper carryway portion of the conveyor. The belt is trained around sprockets (or other reversing structure) mounted on each end of the carryway, and returns along a returnway below the carryway. Such standard conveyor belt modules **101** are known in the art and are not limited to the illustrative flat-top modules. One skilled in the art will recognize that the body of the modules may comprise any suitable configuration, and is not limited to the illustrative embodiment. For example, the body may comprise planar structure that is solid, perforated, “open grid,” flat, shaped or otherwise configured.

(52) In one or more select locations, a dedicated access assembly **200** is inserted between two standard modules **101** to provide a dedicated opening point and-or re-connecting point for the conveyor belt. The dedicated access assembly **200** comprises a series of mating female modules forming a female portion **201** and corresponding male modules **202** forming a male portion. The dedicated access assembly **200** is configured to be opened and-or closed without tools. Alternatively, a single male module and-or a single female module may span the width of the conveyor belt.

(53) In one embodiment, the dedicated access assembly **200** can be a different color from the standard modules and-or contain indicia, such as signs, words or other markings, indicating that the dedicated access assembly **200** is the preferred place to open and-or close the conveyor belt **100** and-or providing instructions for opening and-or closing the conveyor belt.

(54) As shown in FIGS. 2, 3, 5 and 6, the dedicated access assembly **200** can open the conveyor belt by angling the female and male modules **201**, **202** relative to each other into a separation orientation, shown in FIGS. 2 and 5, then unhooking the male and female modules **201**, **202** from each other to open the conveyor belt, as shown in FIGS. 3 and 6. The illustrative male and female modules are configured to only be capable of unhooking and re-hooking when in the angled position.

(55) Referring to FIGS. 7-9, in one embodiment, a dedicated access assembly **200** comprises a series of female modules **201** forming a row of the conveyor belt. Alternatively, a single female module **201** may span the width of the conveyor belt. The female modules each comprise a body having a top surface **210** forming a portion of the top conveying surface, an opposite bottom surface **211** having a drive element **212** extending therefrom, standard hinge elements **220** on a first end of the body for connecting to a standard module **101** (or series of standard modules) in an articulatable manner and a series of open hooks **230** on a second end. The female modules **201** mate with a series of male modules **202** forming another row of the conveyor belt to form a dedicated opening and closing point for the conveyor belt. Alternatively, a single male module can span the width of the conveyor belt. The male modules **202** comprise a body having a top surface

215 forming a portion of the top conveying surface an opposite bottom surface **216** including a drive element **213**, male connecting elements **240** on a first end of the body for mating with the open hooks **230** of the female modules and standard hinge elements **222** on a second end for connecting to standard hinge elements **120** on standard module **101**. The female and male module drive elements **212**, **213** have a pitch equal to the pitch of the standard modules, so that the dedicated access assembly **200** engages a conveyor drive in the same manner as the standard modules.

(56) At the side edges of the row, the female and male portions **201**, **202** have a thickened section **251**, **252**, **253**, **254** that extends in height from the top surfaces **210**, **215** to a bottom surface coextensive with the bottom surface of the drive elements **212**, **213**. The male portion **202** of the dedicated access assembly includes rounded surfaces **255** at the front of thickened edges **252**, **254** configured to abut shaped rear surfaces **256** on the thickened edges **251**, **253** of the female portion **201**. When the modules are flat relative to each other, as shown in FIGS. **8** and **9**, the interference between the surfaces **255**, **256** prevents the male and female rows **201**, **202** from collapsing together.

(57) In the illustrative embodiment, the outside open hooks **2300** are thicker than the interior open hooks **230** in the series, but the invention is not so limited.

(58) The male connecting elements **240** comprise shaped linking elements offset from the open hooks **230** and a trapped rod **242** extending through aligned openings of the shaped linking elements **240**. The illustrative trapped rod is stainless steel, but the invention is not so limited. The thickened side edges **252**, **254** of the male portion **202** trap the trapped rod **242** within the row. Alternatively, the rod **242** may be integrally formed with the male connecting elements **240**.

(59) FIG. **10** is a bottom view of a middle portion of the series of male modules **202**, showing the shaped linking elements **240**. Each illustrative shaped linking element **240** comprises a tapered rounded end **243** configured to abut shaped end edges **262** of the female modules **201** between the open hooks **230** (shown in FIG. **9** and FIG. **12**). The rounded ends **243** prevent the male and female rows from collapsing together. The bottom **244** of the shaped linking elements **240** is recessed to provide clearance for disengaging the open hooks **230** from the trapped rod **242**.

(60) Referring to FIGS. **11** and **12**, the open hooks **230** are configured to hook onto the trapped rod **242** in the spaces between the shaped linking elements **240** of the male module **202**. The open hooks **230** have a round seat **231** sized and dimensioned to seat the trapped rod **242** and allow articulation of the modules relative to each other when the trapped rod **242** is seated in the open hook **230**. An opening **232** to the seat **231** is sized so that the trapped rod **242** can enter and exit the seat at a particular angle, shown as about 45° relative to the top surface **210**. An angled wall **233** forms a surface guiding the rod **242** into the seat **231**.

(61) When the male and female portions **201**, **202** are angled relative to each other, as shown in FIGS. **13** and **14**, a user can manually disengage the open hooks **230** from the trapped rod **242** to open the belt, as shown in FIGS. **15** and **16**. In the illustrative embodiment, the modules are configured to allow disengagement when the modules are bent about 90° relative to each other, but the invention is not so limited. At the selected angle, the shaped edges **256**, **262** of the female portion **201** pass the rounded edges **243**, **255** of the male portion **202**, allowing the rows to collapse together and enabling a user to unhook the female portion **201** from the male portion **202**. When shaped edges **256**, **262** pass a release point **246** at the transition between the round surfaces **255**, **243** and cut-out surface **244** on the bottom of the hinge elements **240**, the rows can collapse together to enable the open hooks **230** to disengage from the trapped rod **242**. The male and female portions may be shaped in any suitable manner to determine the angle at which the portions **201**, **202** can disengage, and the invention is not limited to the illustrative shape and angle.

(62) Referring to FIGS. **17-20**, the illustrative series of female modules **201** comprises a first side module **201a**, a middle module **201b** and a second side module **201c**. The female modules may include alignment tabs **280**, **282** that engage corresponding recesses **281**, **283** for aligning the

modules to form the row. The male portion **202** may also comprises a series of modules connected using alignment tabs or another alignment feature. The illustrative alignment feature may be implemented in other areas of a conveyor belt, not limited to use in a dedicated access assembly. (63) Referring to FIG. **21**, in another embodiment, a dedicated access assembly **300** for a conveyor belt comprises a series of female modules **301** coupled to a series of male modules **302** and one or more latching mechanism **370** for selectively blocking the opening of at least one of the open hooks **330**, shown as the outside open hooks **3300**, to lock the open hooks onto a trapped hinge rod **342** to prevent separation of the modules. The illustrative series of male modules **302** is identical to the series **202** shown in FIGS. **1-20**, but the invention is not so limited. The open hooks **330** are also the same size, shape, configuration and layout as the open hooks **230** of the female portion **201** described above.

(64) Referring to FIG. **22**, the illustrative series of female modules **301** comprises a first side module **301a**, a middle module **301b** and a second side module **301c**. A series of aligned stanchions **380** extends from a bottom surface of each module for receiving an alignment rod **382** to join the modules **301a**, **301b**, **301c** together, maintain their alignment and form a complete row of the conveyor belt.

(65) The illustrative latching mechanism **370**, shown in detail in FIGS. **23** and **24**, comprises a spring clip that is mounted on the alignment rod **382** between two suitably spaced stanchions **380**. FIG. **23** shows the spring clip in a locking position and FIG. **24** shows the spring clip in an open, unlocked position. The spring clip **370** is naturally biased into the locked position of FIG. **23**. The spring clip comprises a body having an opening **372** for receiving the alignment rod **382**, upon which the spring clip **370** can rotate. The body terminates in a tip **374** for selectively blocking the opening of an open hook **3300** and a base **376** that contacts the bottom surface of the module body. A biasing mechanism **378**, shown as a leaf spring, extends between the opening **372** and tip **374**. The leaf spring **378** has an upper span extending from and parallel to the body of the spring clip, a curve near the opening **372** and a bottom span that overlies the bottom surface of the module body. The leaf spring **378** biases the tip **374** away from the bottom surface of the module body, while the base **376** prevents overrotation of the spring clip on the alignment rod **382**. The body of the spring clip can be pushed down, compressing the biasing mechanism.

(66) As shown in FIGS. **25** and **26**, in the natural position, the spring clip **370** blocks the corresponding open hook **3300**, locking the hinge rod **342** in place. To separate the female portion **301** from the male portion **302**, a user depresses the spring clip **370** at each side of the row to open the hooks **3300**, as shown in FIGS. **27** and **28**. Then, the user angles the female portion **301** and male portion **302** relative to each other, as described above and shown in FIGS. **29** and **30**, collapses the rows together and unhooks the female portion **301** from the male portion **302**, as shown in FIGS. **31** and **32**. The spring clip **370** returns to its natural position absent a force upon it. The female portion **301** can clip back onto the male portion **302** to close the conveyor belt, with the action of the hinge rod **342** pushing on the spring clip **370** automatically opening the open hooks **330**.

(67) As shown in FIG. **33**, in another embodiment, a female portion **401** of a dedicated access assembly **400** includes edge modules **401a**, **401c** including spring clips **470** and at least one middle module **401b** that include one or more alignment tabs **480** or other alignment features for aligning the middle module **401b** with the edge modules **401a**, **401c**. The illustrative middle module **401b** also includes a laterally-extending drive element **412** on a bottom surface for driving a conveyor belt including the dedicated access assembly **400**. The illustrative male portion **402** is similar to the male portions **202**, **302** described above and the other features of the female portion are similar to the features of the female portion **302** described above.

(68) Referring to FIG. **34**, in another embodiment, a dedicated access assembly **500** includes mating female and male portions **501**, **502** and a latching mechanism comprising a torsion bar **580** and a lever **570** mounted on the torsion bar **580**. The torsion bar **580** biases the lever **570** into a

locking position, blocking the outside open hook **5300** to prevent disengagement of the female and male portions **501**, **502**.

(69) As shown in FIG. 35, the female portion **501** may comprise a series of aligned modules, including end modules **501a**, **501c** and middle module **501b** that connect to form a row of the conveyor belt. Each end module **501a**, **501c** includes a torsion bar **580** rotatably mounted in and extending between aligned stanchions **582** and a base stanchion **583** in the middle of the module **501a**, **501c**. An alignment rod **584** extends through alignment stanchions **585** on the middle module **501b** and the interior sides of the edge modules **501a**, **501c** and into the base stanchions **583** to connect and align the series of modules together to form a full row of the conveyor belt. Other alignment mechanisms, such as alignment tabs, may alternatively be used.

(70) The torsion bar **580** and associated torsion lever **570** may have any suitable size, shape and configuration. In one embodiment, shown in FIG. 36, the torsion bar **580** has a square mounting end **591** received in a square opening **571** of the torsion lever **570**, so that rotation of the torsion bar **580** also rotates the torsion lever **570**. The base end **592** includes a shaped protrusion **593** that is received in the base stanchion **383** and biases the torsion lever **570** into a locked position, with the tip **572** of the lever **570** trapping a hinge rod in an open hook.

(71) FIGS. 37 and 38 show another embodiment of a conveyor belt **106** employing a dedicate access assembly **600** between a series of standard modules **101** for opening and closing the conveyor belt at a dedicated location between a first row and a second row. The illustrative dedicated access assembly **600** comprises an openable edge portion **601** configured to rotate and snap into and out of engagement with a hinge rod **642** connecting two rows of the conveyor belt together to allow removal of the hinge rod **642** and separation of the two rows. The dedicated assembly **600** could alternatively comprises two rows (or portions of two rows) of non-standard modules.

(72) FIG. 39 is a bottom view of the dedicated access assembly **600** forming a dedicated opening and-or closing point for the conveyor belt **106**. The dedicated access assembly **600** spans three rows of the conveyor belt at a side edge of the conveyor belt. A first edge module **610** connects to other standard modules in the conveyor belt in an articulatable manner using hinge rods, as previously described. A first end of the first edge module **610** connects to the openable edge portion **601** via a standard hinge rod **640**. The first end of the first edge module **610** also connects to an interior conveyor belt module **620** aligned with and inset from the openable edge portion **601** via the standard hinge rod **640**. The interior module **620** and openable edge portion **601** connect to a second edge module **611** via removable hinge rod **642**. The dedicated access assembly **600** further includes a latching mechanism **670** for selectively locking the openable edge portion **601** to the removable hinge rod **642**.

(73) FIGS. 40 and 41 are exploded views of the components of the dedicated access assembly **600**. The illustrative first edge module **610** extends from an inner side edge **612** to a thickened outer side edge **613**, has a flat top conveying surface **614** and an opposite bottom surface from which a drive element **615** extends. A series of standard hinge elements **616** configured to interlace with and connect to a series of standard hinge elements on a standard conveyor belt module extends from a first end. A series of hinge elements **617** extends from the second end, configured to engage hinge elements **606** on the openable edge portion **601**.

(74) The openable edge portion **601** includes a top conveying surface **602** and an opposite bottom surface from which a drive element **605** extends. Articulatable hinge elements **606** extend from the first end and engage the hinge elements **617** of the first edge module **601**, with the hinge rod **640** extending through an aligned passageway formed by hinge elements **617**, **606**. A thickened outer side edge **603** includes a recess **607** at the second end in the bottom surface forming a pocket for trapping the hinge rod **642**. The openable edge portion **601** also includes an open hook **608** at a second end configured to hook onto the hinge rod **642** to lock the openable edge portion **601** to the second edge module **611** and trap the hinge rod **642**. The illustrative open hook **608** snap fits into

and out of engagement with the hinge rod **642**. The recess **607** aligns with the open hook **608**. The thickened outer side edge **603** is spaced from the drive element **605** and includes a protrusion **609** for engaging the latching mechanism **670**, as described below.

(75) The second edge module **611** includes hinge elements **619** for connecting the second edge module **611** to the openable edge portion **601** and the interior module **620**, a thickened outer side edge **623** and hinge elements **624** for connecting to a standard hinge module of the conveyor belt **106**.

(76) The interior module **620** includes a top conveying surface **622** and opposite bottom surface from which a drive element **625** extends, a first set of hinge elements **626** configured to align with hinge elements **617** and **606** and connect via hinge rod **640** and a second set of hinge elements **627** configured to align with the open hook **608**, pocket **607** and hinge elements **619** of the second edge module. Hinge rod **642** extends through the aligned passageway formed by interleaved hinge elements **627**, **619**, open hook **608** and pocket **607**.

(77) In one embodiment, the openable edge portion **601** is formed from a different material than the other belt modules **610**, **611**, **620**. For example, the openable edge portion **601** may be formed of metal or another durable material, while the other belt modules may be plastic.

(78) The latching mechanism **670** is configured to lock the openable edge module **601** onto the hinge rod **642**. The latching mechanism **670** can be opened to allow the openable edge module **601** to rotate about the first hinge rod **640** and out of engagement with the second hinge rod **642**, as shown in FIG. **38**. In the unlatched position, the hinge rod **642** is exposed and can be easily removed, separating the two rows of the conveyor belt **106** held together with the hinge rod **642**. The illustrative latching mechanism **670**, shown in detail in FIG. **42**, comprises a body **671** having a hinge element **672** at a first end for integrating the latching mechanism into the dedicated access assembly **600** by mounting the latching mechanism on the hinge rod **640**. The body **671** includes a biasing mechanism, shown as a spring **673**, and terminates in a tip **674**. A recess **675** in a bottom surface of the body **671** seats the protrusion **609** of the openable edge portion **601**.

(79) As shown in FIG. **43**, in a locked position, the tip **674** of the latching mechanism **670** locks the hinge rod **642** within the trapping pocket **607** and open hook **608**. The spring **673** biases the tip **674** into the locked position, preventing escape of the hinge rod **642**. The hinge element **672** and recess **675** maintain the position of the latching mechanism **670** relative to the openable edge portion **600**.

(80) To open the conveyor belt **106** employing the dedicated access assembly **600**, a user depresses the tip **674** against the biasing mechanism **673** to allow a user to rotate the openable edge portion **601** about hinge rod **640** and out of engagement with the hinge rod **642**. The thus exposed hinge rod **642** can be removed from the conveyor belt to open the conveyor belt. Conversely, a user can lock the openable edge module **601** onto the hinge rod **642** by pressing the hinge rod **642** against the tip **674** to depress the tip **674** against the biasing mechanism **673** and push the hinge rod **642** into the pocket **607** and seat of the open hook **608**.

(81) In another embodiment, shown in FIGS. **45** and **46**, an openable edge module **701** in a dedicated access assembly **700** omits a separate latching mechanism. Instead, a snap fit engagement between the open hook **708** and-or trapping pocket **707** lock the openable edge module **701** in place. In the embodiment of FIGS. **45** and **46**, a drive element **705** on the openable edge module **701** extends to the thickened side edge **703**. A user can manually push the openable edge out of engagement with the open hook **708** and trapping pocket **707** to expose the hinge rod **742** for removal.

(82) The invention has been described relative to certain illustrative embodiments, though those skilled in the art will recognize that modifications may be made without departing from the scope of the invention.

Claims

1. A module for a conveyor belt, comprising: a body extending in height from a top conveying surface to an opposite bottom surface, longitudinally from a first end to a second end and laterally from a first side to a second side; a first set of hinge elements extending from the first end of the body; an open hook extending from the second end of the body for locking the module onto a hinge rod; a recess aligned with the open hook and extending from the second end of the body to form a pocket for trapping the hinge rod, wherein the recess forms part of a thickened outer side edge of the module, and the thickened outer side edge includes the second side of the body; a drive element extending from the bottom surface of the body at the first side of the body and spaced from the thickened outer side edge; and a latching mechanism seated in a space between the thickened outer side edge and the drive element for locking the open hook onto the hinge rod.
2. The module of claim 1, wherein the thickened outer side edge includes a protrusion for positioning the latching mechanism.
3. The module of claim 1, wherein the latching mechanism comprises a spring clip mounted on a rod.
4. The module of claim 3, wherein the spring clip comprises an opening for receiving the rod, a tip for selectively blocking the opening of the open hook and a base that contacts the bottom surface of the module body.
5. The module of claim 1, wherein the module is formed of metal.
6. A module for a conveyor belt, comprising: a body extending in height from a top conveying surface to an opposite bottom surface, longitudinally from a first end to a second end and laterally from a first side to a second side; a first set of hinge elements extending from the first end of the body; an open hook extending from the second end of the body for locking the module onto a hinge rod; and a latching mechanism for locking the open hook onto the hinge rod, the latching mechanism including a tip for locking the hinge rod within the open hook and a biasing mechanism for biasing the tip into a locked position.
7. The module of claim 6, wherein the latching mechanism includes an opening that aligns with the first set of hinge elements.
8. The module of claim 6, further comprising a recess aligned with the open hook and extending from the second end of the body to form a pocket for trapping the hinge rod.
9. The module of claim 6, wherein the latching mechanism includes a positioning recess in a bottom surface and the module body includes a protrusion seated in the positioning recess to maintain the position of the latching mechanism relative to the body.
10. The module of claim 6, wherein the biasing mechanism further includes a leaf spring with an upper span extending parallel to the tip, an opening for receiving a rod, a curve near the opening and a bottom span that overlies the bottom surface of the module body.
11. The module of claim 6, wherein the biasing mechanism comprises a torsion bar and the tip is formed on a lever mounted on the torsion bar.
12. The module of claim 6, wherein the latching mechanism comprises a spring clip mounted on an alignment rod extending between a pair of stanchions.
13. A module for a conveyor belt, comprising: a body extending in height from a top conveying surface to an opposite bottom surface, longitudinally from a first end to a second end and laterally from a first side to a second side; a first set of hinge elements extending from the first end of the body; a pocket extending from the second end of the body for trapping a hinge rod; and a latching mechanism for locking the hinge rod in the pocket, the latching mechanism including a tip for locking the hinge rod within the pocket and a biasing mechanism for biasing the tip into a locked position.
14. The module of claim 13, wherein the latching mechanism includes a hinge element configured to align with the first set of hinge elements.
15. The module of claim 13, further comprising an open hook extending from the second end of the

body and aligned with the pocket.

16. The module of claim 13, wherein the body includes a protrusion and the latching mechanism includes a recess in a bottom surface for seating the protrusion.
